

Teaching Reproducible Research Using R

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Table of contents

About This Book	12
What This Book Represents	13
How This Book Is Organized	14
Reproducibility as a Professional Skill	15
From Assignment to Authorship	16
License	17
How to Cite	17
Version History	17
A Model of Publication in Practice	18
A Note to Readers	19
Course Blueprint	20
Course Philosophy	21
Learning Objectives	22
Course Structure	23
Tuesdays — Concept + Guided Implementation	23
Thursdays — Applied Workshop	23
Assessment Design	24
Weekly Assignments (40%)	24
Midterm Project (30%)	24
Final Project (30%)	25
Conceptual Arc of the Semester	26
Foundations	26
Visualization & Exploration	26
Statistical Inference	26

Reproducible Communication	27
Open Science Integration	28
From Coursework to Authorship	29
Institutional Policies & Resources	30
Closing Note	31
Assignment 1: Base R Foundations	32
Overview	32
Learning Objectives	33
Textbook Connection	34
Submission Instructions	35
Assignment Tasks	36
1. R as a Calculator	36
2. Vectors	36
3. Data Frames	36
4. Built-in Datasets	37
Reflection	37
Reproducibility Practice	38
Assignment 2: Exploring Pixar with the Tidyverse	39
Overview	39
Learning Objectives	39
Textbook Connection	40
Submission Instructions	40
Assignment Tasks	41
1. Exploring Pixar Genres	41
1.1 What Genres Appear in Pixar Films?	41
1.2 How Many Genres Does Each Film Have?	41
2. Which Pixar Films Made the Most Money?	42
3. Turning <code>cinema_score</code> Grades into Numbers	43
4. Pixar Through the Decades	43
4.1 Create a Decade Variable	43
4.2 Summarize Pixar Films by Decade	44
5. Pixar Data Challenge	44

Reflection	46
Reproducibility Practice	47
Assignment 3: NYPD Shooting Incidents — Cleaning, Insights, and Visualization	48
Overview	48
Learning Objectives	49
Textbook Connection	50
Submission Instructions	51
Assignment Tasks	52
0. Data Source	52
1. Cleaning and Feature Engineering	52
1A. Missing data removal (defensible choice)	52
1B. Lowercase a column	52
1C. Create <code>time_of_day</code>	53
1D. Create <code>days_since</code>	53
2. Insights	53
3. Visualizations (ggplot2)	53
Plot 3A. <code>time_of_day</code> plot	53
Plot 3B. personal insight plot	54
4. Written Summary	54
Starter Code Template	54
Reproducibility Practice	57
Assignment 4: Merging Data and Comparing Means with t-tests	58
Overview	58
Learning Objectives	59
Textbook Connection	60
Submission Instructions	61
Assignment Tasks	62
1. Data Wrangling (Import + Merge)	62
2. Handling Missing Data (Imputation)	63
3. Dependent t-test (Paired)	63
4. Independent t-tests (Sex + Department)	63
4A. Add a Sex column (using <code>babynames</code>)	63
4B. Improvement t-test by Sex	64

4C. Improvement analysis by Department	64
Written Summary	64
Reproducibility Practice	65
Assignment 5: R Script → Quarto Report	66
Overview	66
Learning Objectives	67
Textbook Connection	68
Submission Instructions	69
Assignment Tasks	70
1. YAML header	70
2. Setup chunk	70
3. Minimum required code chunks (4)	70
4. Narrative text	71
5. Inline R expression	71
6. Hyperlink to the dataset	71
Reflection	71
Publishing Instructions	72
Reproducibility Practice (Quarto Reporting)	73
Publishing Instructions	74
Assignment 6: ANOVA and Payment Patterns in NYC Camera Violations	75
Overview	75
Learning Objectives	76
Textbook Connection	77
Submission Instructions	78
Assignment Tasks	79
1. Data Ingest	79
2. Data Cleaning and Feature Engineering	79
2A. Numeric Conversion	79
2B. Date Cleaning	80
2C. Time Cleaning	80
3. Analysis 1: Day of Week and Fine Amount	81

4. Analysis 2: Time of Day and Fine Amount	81
5. Analysis 3: Violation Type and Fine Amount	82
6. Analysis 4: Violation Type and Payment Amount	82
Final Reflection	83
Reproducibility Practice	84
Publishing Instructions	85
Midterm: Exercise & Sleep	86
Overview	86
Learning Objectives	87
Textbook Connection	88
Submission Instructions	89
Midterm Tasks	90
1. Setup & Data Import	90
Tasks	90
2. Merge & Base Cleaning	90
Tasks	90
Requirements	91
3. Create Derived Variables	91
Tasks	91
Requirements	91
4. Descriptive Statistics	91
Tasks	91
Requirements	92
5. Visualizations (3 plots)	92
Requirements	92
6. t-tests (Two)	93
Requirements	93
7. ANOVAs (Two) + Post-hoc Tests	93
Requirements	93
8. Synthesis & Recommendation	94
Requirements	94
9. Reflection	94
Reproducibility Practice	95
Publishing Instructions	96

Assignment 7: NBA Analytics — Exploring Team Performance Through Reproducible Analysis	97
Overview	97
Learning Objectives	98
Textbook Connection	99
Submission Instructions	100
Assignment Tasks	101
1. Data Ingest and Preparation	101
Required Tasks	101
Requirements	102
2. Adding Conference Information	102
3. Visual Exploration	102
Required Plot 1	103
Required Plot 2	103
4. Correlation Analysis	103
A. Point-Biserial Correlation	103
B. Correlation Matrix	104
C. Partial Correlation	104
5. Executive Memo to Adam Silver	105
Guidelines for Interpretation	106
Reflection	107
Reproducibility Practice	108
Publishing Instructions	109
Assignment 8: Florida Crime Analytics — Uncovering the Root of Florida’s Crime Surge	110
Overview	110
Learning Objectives	111
Textbook Connection	112
Textbook Connection	113
Submission Instructions	114

Assignment Tasks	115
1. Data Loading and Preparation	115
Required Tasks	115
2. Exploratory Data Analysis	116
Required Components	116
Bonus (Optional)	116
3. Correlation Analysis	116
Required Tasks	116
4. Building Regression Models	117
Required Models	117
Model Comparison	117
5. Executive Memo to the Chief of the Florida Police Department	118
Guidelines for Interpretation	119
Final Question	120
Reflection	121
Reproducibility Practice	122
Publishing Instructions	123
Assignment 9: Streaming Analytics — Understanding Platform Popularity Across Age Groups	124
Overview	124
Learning Objectives	125
Textbook Connection	126
Submission Instructions	127
Assignment Tasks	128
1. Data Preparation	128
Required Tasks	128
2. Visualization	128
Required Visualization 1	128
Required Visualization 2	129
3. Chi-Square Test of Independence	129
4. Observed, Expected, and Residual Values	129
5. Contributions to the Chi-Square Statistic	130
6. Effect Size (Cramer's V)	130
7. Final Interpretation	131

Deliverables	132
Guidelines for Interpretation	133
Reflection	134
Reproducibility Practice	135
Publishing Instructions	136
Assignment 10: Wage Analytics — Predicting High vs Low Earners Using Logistic Regression	137
Overview	137
Learning Objectives	138
Textbook Connection	139
Submission Instructions	140
Assignment Tasks	141
1. Create the WageCategory Variable	141
Required Tasks	141
Important	141
2. Data Cleaning	141
Required Tasks	142
3. Classical Statistical Tests	142
A. t-test	142
B. ANOVA	142
C. Chi-Square Test	143
4. Logistic Regression Model	143
Train/Test Split	143
Model Specification	144
5. Model Evaluation on Test Data	144
ROC Curve and AUC	145
6. Final Interpretation	145
Guidelines for Interpretation	146
Reflection	147
Reproducibility Practice	148
Publishing Instructions	149

Final Project: NYC Open Data 2026 Proposal	150
Overview	150
Learning Objectives	151
Textbook Connection	152
Project Requirements	153
1. Dataset Selection	153
2. Reproducible Analysis (Quarto)	153
3. Proposal-Style Structure	154
A. Title & Event Description	154
B. Dataset(s) Used	154
C. Analysis	155
D. Visualizations	155
E. Audience & Relevance	155
F. Connection to Open Data	156
Length & Formatting Requirements	157
Class Presentation	158
Class Publication (Optional but Encouraged)	159
Grading (100 Points Total)	160
Reproducibility Practice	161
Resources	162
Submission	163
Takeaway	164
Final Assignment: Building Your Reproducible Quarto Book	165
Overview	165
Learning Objectives	166
Textbook Connection	167
Assignment Requirements	168
1. Create a Quarto Book Project	168
2. Required Chapters (In This Order)	168
3. Title Page	169

4. Cover Image	169
5. Chapter Introductions	170
6. Figure and Table Captions	170
7. Final Reflection Chapter	170
Book Quality Standards	172
Publishing Requirement	173
Grading (100 Points Total)	174
Reproducibility Practice	175
Takeaway	176
Appendices	177
References	177
References	178

About This Book

This book represents the applied companion to [Reproducible Research Using R](#).

Over the course of the semester, students transformed structured assignments into fully reproducible analyses using real datasets, transparent workflows, and public-facing communication. What began as individual homework submissions evolved into polished research chapters assembled into this cohesive volume.

This book treats reproducibility not as an enhancement, but as a default professional practice.

Each chapter reflects a progression of skills: data import, cleaning, visualization, statistical testing, modeling, interpretation, and communication. More importantly, each chapter reflects a shift in mindset — from “running code” to designing complete analytical workflows.

This is not just a collection of assignments.
It is a curated portfolio of applied reproducible research.

What This Book Represents

This volume showcases:

- Applied statistical reasoning
- Transparent data cleaning and transformation
- Reproducible modeling and interpretation
- Clear, professional communication of results
- Public-facing research using real-world datasets

Every analysis in this book can be rerun from start to finish. All figures, tables, and results are generated directly from executable code. Nothing has been manually edited or pasted from external software.

The goal is not perfection. The goal is transparency.

How This Book Is Organized

The chapters follow the structure of the course and mirror the conceptual arc of *Reproducible Research Using R*.

- [Base R Foundations](#)
- [mtcars Wrangling and Feature Engineering](#)
- [NYPD Shooting Incidents — Cleaning, Insights, and Visualization](#)
- [Merging Data and Comparing Means with t-tests](#)
- [R Script → Quarto Report](#)
- [ANOVA and Payment Patterns in NYC Camera Violations](#)
- [Midterm: Exercise & Sleep – Experimental Analysis and Inference](#)
- [NBA Analytics – Correlation and Partial Correlation](#)
- [Florida Crime Analytics – Regression Modeling and Model Comparison](#)
- [Streaming Analytics – Categorical Analysis and Chi-Square Testing](#)
- [Wage Analytics – Logistic Regression and Predictive Modeling](#)
- [Final Project – Independent Civic Data Research](#)
- [Final Assignment – Building Your Reproducible Quarto Book](#)

Each chapter applies a specific methodological framework to a real dataset.

Together, they form a complete analytical portfolio.

Reproducibility as a Professional Skill

Throughout this book, reproducibility is treated as a default practice rather than an optional enhancement.

Each chapter:

- Loads data programmatically
- Cleans variables transparently
- Documents transformation decisions
- Avoids hard-coded statistical results
- Produces outputs directly from executable code

This workflow ensures that analyses are inspectable, repeatable, and defensible — qualities expected in graduate research, industry analytics, and public-facing data work.

From Assignment to Authorship

By assembling these projects into a Quarto book, students move beyond submission-based coursework and into publication-based thinking.

This book serves as:

- A professional portfolio artifact
- A demonstration of applied statistical competency
- Evidence of reproducible workflow mastery
- A foundation for conference submissions and civic engagement

Completing this book is not simply a course requirement.

It is a transition point — from student to analytical author.

License

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How to Cite

If you use this textbook in your teaching, research, or projects, please cite it as:

Martinez, C. (2026). *Teaching Reproducible Research Using R* (Version 1.0.1). Zenodo.
<https://doi.org/10.5281/zenodo.19139302>

Version History

v1.0.1 (2026)

- Initial public release
 - Full textbook curriculum covering data analysis, visualization, and modeling in R
 - Used in Fall 2025 coursework at Brooklyn College (CUNY)
-

A Model of Publication in Practice

A previous cohort of students transformed their final projects into a collective publication:
[NYC Open Data Student Gallery - Brooklyn College](#)

A Note to Readers

If you are a future employer, graduate program, research lab, or collaborator reviewing this book, you are not just reading assignments — you are reviewing a semester-long demonstration of analytical growth.

If you are a student currently enrolled in the course, this book represents what is possible when structured workflow and intentional practice are sustained over time.

Reproducibility is not a feature of advanced researchers.
It is a habit formed early.

Course Blueprint

Semester: Fall 2025

Instructor: Christian A. Martinez

Institution: Brooklyn College, CUNY

Course Philosophy

This course is structured around a simple premise: reproducibility is not an advanced skill — it is a foundational habit.

Rather than treating R as a programming language to learn in isolation, this course approaches R as a complete research environment. Students learn how to move from raw data to interpretable results using transparent, inspectable, and *reproducible* workflows.

By the end of the semester, students do not just complete assignments — they produce a cohesive analytical portfolio.

Learning Objectives

By the end of this course, students are able to:

- Write and execute R scripts for applied data analysis
 - Clean and wrangle datasets using reproducible workflows
 - Create publication-quality visualizations using `ggplot2`
 - Conduct and interpret common statistical analyses (t-tests, ANOVA, regression, logistic regression, Chi-Square)
 - Reproduce analyses from published psychology papers with open datasets
 - Communicate results in structured Quarto documents
 - Apply open science practices (e.g., OSF, GitHub, transparent documentation)
 - Assemble individual analyses into a polished Quarto book
-

Course Structure

The course follows a consistent weekly rhythm designed to reinforce applied learning.

Tuesdays — Concept + Guided Implementation

- Short conceptual lecture (20–25 minutes)
- Guided mini-lab exercises
- Emphasis on understanding the logic behind each method

Thursdays — Applied Workshop

- Live coding demonstration
- Hands-on work with real datasets
- Reproduction of analyses from published psychology papers
- Debugging as a collaborative process
- Screen-sharing and peer problem solving encouraged

This structure ensures that students immediately apply conceptual material in authentic research contexts.

Assessment Design

Evaluation in this course is cumulative and portfolio-driven.

Weekly Assignments (40%)

Each week introduces a new methodological framework. Assignments require students to:

- Clean and analyze real datasets
- Interpret statistical output
- Produce reproducible visualizations
- Write structured narrative explanations

Assignments progressively increase in complexity.

Midterm Project (30%)

The midterm integrates:

- Data import and merging
- Variable engineering
- t-tests and ANOVA
- Post-hoc comparisons
- Evidence-based recommendation

Students demonstrate their ability to conduct a complete inferential workflow independently.

Final Project (30%)

Students design and conduct an original research project using NYC Open Data or other civic datasets.

The final submission includes:

- Fully reproducible analysis
 - Clear research framing
 - Professional visualizations
 - Public-facing interpretation
 - In-class presentation
-

Conceptual Arc of the Semester

The course progresses through four major phases:

Foundations

- R basics
- Data structures
- Functions and packages
- Tidyverse workflows

Visualization & Exploration

- `ggplot2`
- Feature engineering
- Exploratory data analysis
- Data cleaning decisions

Statistical Inference

- t-tests
- ANOVA
- Correlation
- Linear regression
- Chi-Square tests
- Logistic regression

Reproducible Communication

- Quarto documents
- Structured reporting
- Transparent code practices
- Building a Quarto book

Each stage builds directly on the previous one.

Open Science Integration

Reproducibility is treated as a default expectation.

Students are required to:

- Load data programmatically
- Avoid manual spreadsheet edits
- Document all cleaning decisions
- Avoid hard-coding statistical results
- Use structured document formats
- Present analyses transparently

Open science practices such as preregistration, OSF usage, and GitHub version control are introduced and encouraged throughout the semester.

From Coursework to Authorship

The final assignment requires students to assemble all homeworks into a cohesive Quarto book, moving away from “one-and-done” assignments.

This transforms:

- Discrete homework submissions
into
- A structured analytical portfolio

The result is a professional artifact suitable for:

- Graduate school applications
- Research lab submissions
- Industry analytics portfolios
- Conference presentations
- Public-facing civic research

The structure of the course ensures that students leave not only with technical fluency in R, but with a coherent body of *reproducible* work.

Institutional Policies & Resources

The course follows all Brooklyn College and CUNY policies regarding academic integrity, accessibility services, and student support resources.

Students are expected to adhere to the CUNY Academic Integrity Policy and to utilize campus support services when needed.

For official policy details, students should refer to the Brooklyn College website and Undergraduate Bulletin.

Closing Note

Reproducible research is not about perfection.

*It is about **clarity**.*

*It is about **transparency**.*

*It is about building **analytical habits** that scale beyond a single course.*

This blueprint reflects a semester-long progression from foundational skills to independent analytical authorship.

Assignment 1: Base R Foundations

Overview

This assignment introduces core concepts in base R, including arithmetic operations, object assignment, vectors, data frames, and built-in datasets. The goal is to build comfort with fundamental R syntax before moving into more advanced workflows.

You will complete all tasks in a single `.R` script.

Learning Objectives

By completing this assignment, you will be able to:

- Perform arithmetic operations in R
 - Assign results to objects using `<-`
 - Create and manipulate vectors
 - Construct and subset data frames
 - Explore built-in datasets
 - Produce a simple base R visualization
 - Organize a script using clear comments
-

Textbook Connection

This assignment builds directly from **Chapter 1: Introduction to R** in [Reproducible Research Using R](#).

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

Submission Instructions

Submit **one .R script file**.

Your script must:

- Be clearly organized into labeled sections
(*e.g., # Part 1: R as a Calculator*)
 - Include all required calculations and outputs
 - Include your reflection as comments at the end
 - Run from top to bottom without errors
-

Assignment Tasks

1. R as a Calculator

1. Calculate $12 * 2$.
 2. Calculate the square root of 144.
 3. Calculate $100 / 7$ and round to **2 decimals**.
 4. Save the result of $15 * 4$ as `my_product`.
 5. Save $100 / 7$, rounded to **5 decimals**, as `my_second_product`.
-

2. Vectors

1. Create a numeric vector of five random numbers named `randy`.
 2. Compute the mean, median, minimum, and maximum of `randy`.
 3. Create another numeric vector of five random numbers named `randy2`.
 4. Create `randy_combined` by adding `randy` and `randy2`.
 5. Create a character vector of five favorite movies.
 6. Use indexing to print the fourth movie.
 7. Create a factor with 11 entries of favorite foods (repeat some).
 8. Use `table()` to count frequencies.
 9. Sort the table in descending order and save as `organized_table`.
-

3. Data Frames

1. Create a data frame called `class_df` with columns:
 - name

- age
 - major
2. Add at least four fictional classmates.
 3. Create `age_in_2009`.
 4. Create `favorite_professor`.
 5. Use indexing to:
 - Select only name and major.
 - Filter rows for Psych majors.
-

4. Built-in Datasets

Using `mtcars`:

1. Display the first 11 rows.
 2. List all column names.
 3. Run `str(mtcars)` and describe column types in a comment.
 4. Run `summary(mtcars)`.
 5. Create a base R scatterplot titled:
“My First Scatterplot”
-

Reflection

At the end of your script, answer in 2–3 sentences (as comments):

What was the most surprising or confusing thing you learned about R so far?

Reproducibility Practice

For this first assignment, focus on script organization and documentation.

At the top of your script, include a structured comment block containing:

- Assignment name
- Your name
- Date
- A brief description of what the script accomplishes

Your script should:

- Be divided into clearly labeled sections
- Run from top to bottom without manual intervention
- Avoid unnecessary objects

This assignment establishes your baseline reproducibility habits for the semester.

Assignment 2: Exploring Pixar with the Tidyverse

Overview

This assignment continues our introduction to the **Tidyverse** using data from the `pixarfilms` package.

The `pixarfilms` package contains several datasets describing Pixar movies, including their genres, box-office performance, audience responses, and release information.

You will use these datasets to answer a series of questions about Pixar's history and films.

Learning Objectives

By completing this assignment, you will be able to:

- Use `dplyr` verbs such as `select()`, `distinct()`, `mutate()`, `count()`, `group_by()`, `summarize()`, and `arrange()` to explore data
 - Identify unique values within a dataset
 - Create new variables from existing variables
 - Use `case_when()` to recode categorical information
 - Identify the highest and lowest values using `slice_max()` and `slice_min()`
 - Calculate grouped summary statistics
 - Apply Tidyverse skills to a new dataset independently
 - Develop and answer your own data-driven question
-

Textbook Connection

This assignment builds directly from **Chapter 2: Introduction to tidyverse** in *Reproducible Research Using R*.

Students are encouraged to review the chapter and the code we worked through together in class before beginning this assignment.

Submission Instructions

Submit **one .R script file**.

Your script must:

- Run from top to bottom without errors
- Include clear section comments for each question
- Save newly created datasets when instructed
- Contain your written responses as comments where requested

Begin your script by loading the packages needed for this assignment:

```
library(tidyverse)
library(pixarfilms)
```

Assignment Tasks

1. Exploring Pixar Genres

The `genres` dataset contains the genres associated with each Pixar film. Notice that a movie can appear more than once because a single movie may belong to multiple genres.

1.1 What Genres Appear in Pixar Films?

First, let's find out which genres appear in the dataset.

Using the `genres` dataset:

1. Use `distinct()` to identify the unique genres represented in Pixar films.
2. Determine the **total number of distinct genres**.
3. Arrange the unique genres alphabetically.

Question: How many distinct genres appear in the dataset?

1.2 How Many Genres Does Each Film Have?

Next, investigate how many genres are associated with each Pixar film.

Using the `genres` dataset:

1. Count the number of genres associated with each film.
2. Arrange the results from the **greatest number of genres to the fewest**.
3. Save the result as `film_genres`.

Question: Which Pixar film or films are associated with the greatest number of genres?

2. Which Pixar Films Made the Most Money?

The `box_office` dataset contains information about the production budget and box-office performance of Pixar films.

For this assignment, we will estimate how much money a film made by subtracting its production budget from its worldwide box-office revenue.

Using `box_office`:

1. Use `select()` to keep only the following variables:
 - `film`
 - `budget`
 - `box_office_worldwide`
2. Use `mutate()` to create a new variable called `net`, calculated as:
`worldwide box office - budget`
3. Save the resulting dataset as `pixar_money`.
4. Using `slice_max()`, identify the **three films with the highest net**.
5. Using `slice_min()`, identify the **three films with the lowest net**.

Questions:

- Which Pixar film had the highest estimated net?
- Which Pixar film had the lowest estimated net?

Note

For this assignment, `net` is a simplified measure of financial performance. A film's production budget and worldwide box-office revenue do not account for all costs or sources of revenue.

3. Turning `cinema_score` Grades into Numbers

The `public_response` dataset contains several measures of how Pixar films were received by critics and audiences.

One of these variables is `cinema_score`, which uses letter grades such as "A+", "A", "A-", and so on.

Let's convert these letter grades into numeric scores.

Using the `public_response` dataset:

1. Use `mutate()` and `case_when()` to create a new variable called `cinema_numeric`.
2. Use the **grading scale from our course syllabus** to determine the numeric value associated with each letter grade.
3. Save the resulting dataset as `pixar_scores`.
4. Calculate the **average numeric `cinema_score`** across Pixar films.
5. Identify the **three films with the highest numeric `cinema_score`**.
6. Identify the **three films with the lowest numeric `cinema_score`**.

Questions:

- What was the average numeric `cinema_score` for Pixar films?
 - Which films received the highest converted scores?
 - Which films received the lowest converted scores?
-

4. Pixar Through the Decades

The `pixar_films` dataset contains basic information about Pixar's films, including their release dates and runtimes.

We want to investigate how Pixar's films have changed across decades.

4.1 Create a Decade Variable

Using `pixar_films`, use `mutate()` to create a new variable called `decade` based on `release_date`.

Your categories should identify films as belonging to the:

- 1990s
- 2000s

- 2010s
- 2020s

Save the resulting dataset as `pixar_decades`.

4.2 Summarize Pixar Films by Decade

Next, group the films by `decade` and calculate:

- Number of films
- Mean runtime
- Median runtime
- Minimum runtime
- Maximum runtime

Arrange the resulting table from the **most recent decade to the oldest**.

Save this summary as `decade_summary`.

Question: Based on your summary, do Pixar movies appear to have become longer, shorter, or stayed approximately the same length over time? Briefly explain your answer as a comment in your script.

5. Pixar Data Challenge

Now it's your turn!

Using **any dataset in the `pixarfilms` package**, come up with **one question about Pixar that you find interesting**.

Some possibilities include questions about:

- Box-office performance
- Genres
- Movie ratings
- Academy Awards
- Release dates
- Runtime

You are not limited to these topics.

Write your question as a comment in your script.

Then:

1. Use at least **three Tidyverse functions** to investigate your question.
 2. Show the code used to answer it.
 3. Write **1–2 sentences as comments** explaining what you found.
-

Reflection

At the end of your script, answer the following questions as comments:

- Which part of this assignment felt the hardest for you? Why?
- Which part felt the easiest for you? Why?
- Which Pixar film is your favorite? (*There is no wrong answer... probably.*)

Reproducibility Practice

This assignment focuses on creating a **clear and reproducible data-analysis workflow**.

In your script:

- Use section headers that match Questions 1–5.
- Give newly created objects clear and meaningful names.
- Do not overwrite the original datasets from the `pixarfilms` package.
- Use `View()`, `glimpse()`, `summary()`, `count()`, or other checkpoint functions when helpful to inspect your work.
- Make sure your completed script runs from **top to bottom without errors**.

Remember: the goal is not simply to get an answer. Your code should make it possible for someone else to understand **how you got there**.

Assignment 3: NYPD Shooting Incidents — Cleaning, Insights, and Visualization

Overview

In this assignment, you will work with **NYPD Shooting Incident Data** using the `nycOpenData` package to practice data cleaning, feature engineering, exploratory insight generation, and data visualization with `ggplot2`.

Compared to previous assignments, this one is more open-ended: you will make a few analytic decisions (e.g., how to define time-of-day categories and which variables to explore for your second insight). This marks your transition from structured exercises to defensible analytic decision-making.

Learning Objectives

By completing this assignment, you will be able to:

- Retrieve real-world NYC civic data using `nycOpenData`
 - Identify and handle missing data in a defensible way
 - Engineer new features from dates/times and categorical columns
 - Generate exploratory insights using grouping and summaries
 - Communicate findings using well-designed `ggplot2` visualizations
 - Write a short interpretation that connects summaries and plots to your insights
-

Textbook Connection

This assignment builds directly from **Chapter 3: Visualizations** in [Reproducible Research Using R](#).

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

Submission Instructions

Submit **one .R script file**.

Your script must:

- Run from top to bottom without errors
 - Include your insight statements and written summary as **comments** where requested
 - Contain **two ggplot2 plots** with clear titles and labeled axes
 - Include at least **one faceted plot** using `facet_wrap()`
-

Assignment Tasks

0. Data Source

You must use the `nycOpenData` package and the function:

```
nyc_shooting_incidents()
```

1. Cleaning and Feature Engineering

Complete all of the following in your script:

1A. Missing data removal (defensible choice)

- Choose **one column** where it is reasonable to remove missing values.
- Compute and report (in a comment) **how many values are missing** in that column before removal.
- Remove rows with missing values in that column.
- In a comment, explain **why** that column was “safe” to use.

1B. Lowercase a column

- Choose one column (e.g., `borough`, `perp_race`, `vic_race`) and convert values to lowercase.

1C. Create `time_of_day`

- Create a new column called `time_of_day` with three categories:
 - Morning
 - Afternoon
 - Night
- You define the time boundaries.
- Briefly note your boundaries in a comment.

1D. Create `days_since`

- Create a column called `days_since` showing the number of days between today's date and the date of the shooting.
 - Use `as.Date()` and/or `lubridate` as needed.
-

2. Insights

Provide the following insights as **comments** in your script:

1. **One insight about `time_of_day`**
Example format: "Shootings are more frequent during ____ than ____ suggestion..."
 2. **One additional insight of your choice**
Explore anything you find interesting in the dataset (be creative).
-

3. Visualizations (ggplot2)

Create two plots using `ggplot2`:

Plot 3A. `time_of_day` plot

- Must directly relate to your `time_of_day` variable

Plot 3B. personal insight plot

- Must support your second insight

Both plots must include:

- Color using an aesthetic mapping (`aes(color = ...)` or `aes(fill = ...)`)
- Informative axis labels
- A clear title
- Custom font/size settings via `theme()`

Additional requirement:

- At least **one** plot must use `facet_wrap()`
-

4. Written Summary

At the end of your script, write a short paragraph (as comments) describing:

- What you discovered from your insights
 - What the graphs revealed that was not obvious from the raw data
-

Starter Code Template

Copy this template into your script and complete the tasks above. (Wickham (2023)) (Spinu et al. (2024)) (Martinez (2026))

```
#####  
# Assignment 3: NYPD Shootings  
# Cleaning, Insights, and Visualization  
#####  
  
# ---- Setup ----  
library(tidyverse)  
library(lubridate)  
library(nycOpenData)
```

```

# ---- Data Ingest ----
shooting_data <- nyc_shooting_incidents()

# =====
# 1. CLEANING & FEATURE ENGINEERING
# =====

# 1A. Missingness (choose one column to drop NAs)
# missing_n <- sum(is.na(shooting_data$YOUR_COLUMN))
# Write a comment: how many were missing + why this column is safe
# cleaned <- shooting_data %>% drop_na(YOUR_COLUMN)

# 1B. Lowercase one categorical column
# cleaned <- cleaned %>% mutate(YOUR_COLUMN = str_to_lower(YOUR_COLUMN))

# 1C. Create time_of_day (Morning / Afternoon / Night)
# cleaned <- cleaned %>%
#   mutate(
#     occur_date = as.Date(occur_date),
#     occur_time = hm(occur_time),
#     hour = hour(occur_time),
#     time_of_day = case_when(
#       hour >= ___ & hour < ___ ~ "Morning",
#       hour >= ___ & hour < ___ ~ "Afternoon",
#       TRUE ~ "Night"
#     )
#   )

# 1D. Create days_since
# cleaned <- cleaned %>%
#   mutate(days_since = as.integer(Sys.Date() - as.Date(occur_date)))

# =====
# 2. INSIGHTS (write as comments)
# =====

# =====
# 3. VISUALIZATIONS
# =====

```

```
# Plot A: time_of_day plot (include facet_wrap on at least one plot)

# Plot B: personal insight plot

# =====
# 4. WRITTEN SUMMARY (comments)
# =====
```

Reproducibility Practice

This assignment focuses on reproducibility when working with **real civic data** and **date/time feature engineering**.

In your script:

- Clearly document your analytic choices (especially your `time_of_day` boundaries) in comments.
- Keep your pipeline modular by creating a cleaned object (e.g., `cleaned`), and build from it.
- After creating your new variables, include a quick checkpoint such as:
 - `count(cleaned, time_of_day)` or
 - `summary(cleaned$days_since)` to verify your features were created correctly.

The goal is that another reader can see *exactly what choices you made*, reproduce your engineered variables, and understand how you arrived at your two insights.

Assignment 4: Merging Data and Comparing Means with t-tests

Overview

This assignment builds on our lesson about **merging datasets**, **handling missing data**, and running **t-tests** in R. You will work with two files:

- **Employee Info** (Excel): first name, last name, department (30 employees)
- **Employee Scores** (CSV): a single **Name** column plus two test scores (**Score Before**, **Score After**)

Your goal is to merge the datasets, clean them, and answer questions using **t-tests**.

Learning Objectives

By completing this assignment, you will be able to:

- Import data from both Excel and CSV formats into R
 - Resolve key mismatches across datasets (split/unite names to form a join key)
 - Merge datasets using a defensible join strategy
 - Diagnose and handle missing values in numeric columns
 - Run and interpret:
 - a **paired (dependent) t-test**
 - an **independent t-test**
 - Create a derived variable (**Improvement**) and use it in statistical comparisons
 - Communicate results clearly (test statistic, p-value, interpretation)
-

Textbook Connection

This assignment builds directly from **Chapter 4: Comparing Two Groups: Data Wrangling, Visualization, and t-Tests** in [Reproducible Research Using R](#).

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

Submission Instructions

Submit **one .R script file**.

Your script must:

- Run from top to bottom without errors
 - Include clear section comments separating each part of the assignment
 - Include your interpretations and written summary as **comments** where requested
 - Load all packages used via `library(...)`
-

Assignment Tasks

1. Data Wrangling (Import + Merge)

1. Import both datasets into R:
 - Employee Info (Excel)
 - Employee Scores (CSV)
2. Notice the mismatch:
 - one dataset has `First Name + Last Name`
 - the other has a single `Name` column
3. Create a matching join key by doing **one** of the following:
 - **Option A:** split `Name` into first/last, or
 - **Option B:** unite first/last into a single full name
 - **Clean the names:** make sure all column names are lowercase and are connected through an `_` (e.g. `score_before`)

 When cleaning column names:

The `clean_names()` function from the `janitor` can be very helpful.

4. Rename columns if needed so the join key matches in both datasets.
5. Merge the datasets so each employee has:
 - department
 - first name and last name (or full name)
 - Score Before and Score After
6. After merging, you should have **30 rows**.
 - Write a short comment confirming the row count and what join you used.

2. Handling Missing Data (Imputation)

Some `Score Before` or `Score After` values have been removed on purpose.

1. Calculate the average of each score column:
 - `Score Before`
 - `Score After`
 2. Replace missing values in each column with that column's average.
 - *Tip:* `mean(..., na.rm = TRUE)` and `replace_na()` (or base R) will be useful.
 3. Write a comment confirming how many missing values were replaced in each column.
-

3. Dependent t-test (Paired)

1. Run a **paired t-test** comparing `Score Before` vs `Score After`.
 2. Report (as comments):
 - test statistic
 - p-value
 - a 1–2 sentence interpretation (Did performance change after the intervention?)
-

4. Independent t-tests (Sex + Department)

4A. Add a Sex column (using `babynames`)

1. Add a `Sex` column by merging on first names using the `babynames` package.
2. If a name does not have an associated sex, remove it from the dataset.
3. Write a short comment reporting how many rows remain after this filtering step.

4B. Improvement t-test by Sex

1. Create a new column:

```
Improvement = score_after - score_before
```

2. Run an **independent t-test** comparing **Improvement** between male and female employees.
3. Report (as comments):
 - test statistic
 - p-value
 - interpretation

4C. Improvement analysis by Department

Using the same **Improvement** column, run a t-test in conjunction with the **department** column.

You can do this in one of two ways:

- If department has **two groups**, run a standard independent t-test:
 - `t.test(Improvement ~ department, data = ...)`
- If department has **more than two groups**, run **multiple** pairwise t-tests (or clearly explain why a t-test is not appropriate and what you would use instead).

Write a few sentences summarizing what you found.

Written Summary

At the end of your script, write a short paragraph (as comments) describing:

- Your overall findings across the t-tests
- Any difficulties you experienced during merging, missing data handling, or interpretation

Reproducibility Practice

This assignment focuses on reproducible workflows for **joining datasets** and **handling missing data**, where mistakes can silently change results.

In your script:

- Print or comment the number of rows **before and after** the merge (confirm you ended with 30 rows).
- Explicitly document which join you used (e.g., `left_join`) and *why*.
- After replacing missing values, include a “checkpoint” that confirms there are no remaining NAs in the score columns (e.g., `sum(is.na(score_before))`).
- Keep key variables named consistently (e.g., `score_before`, `score_after`, `improvement`) so your analysis is easy to verify.

The goal is that another reader could review your script and confirm the merge logic, the imputation logic, and the statistical results without guessing.

Assignment 5: R Script → Quarto Report

Overview

This week, you will take your Week 3 homework assignment (NYPD Shooting Incidents) and convert it into a **Quarto** report (`.qmd`). Using the same prompt and the work you already completed, your task is to create a fully reproducible, publishable HTML report.

Your final output should read like a short research note: code + narrative explanations + tables/plots + brief interpretation.

Learning Objectives

By completing this assignment, you will be able to:

- Convert an analysis workflow from an R script into a reproducible Quarto document
 - Organize work into labeled code chunks that render cleanly to HTML
 - Combine narrative text with analysis code to communicate findings
 - Use inline R expressions to report values dynamically
 - Create publication-quality tables and visualizations for the web
 - Publish a reproducible report and share it via a link
-

Textbook Connection

This assignment builds directly from **Chapter 10: Reproducible Reporting** in [Reproducible Research Using R](#).

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

Submission Instructions

Submit **two things**:

1. A link to your published HTML report (e.g., RPubS, Posit Connect/Cloud, GitHub Pages, etc.)
2. Your `.qmd` file

Your Quarto report must render to HTML without errors.

Assignment Tasks

Your Quarto document must include the following sections and components.

1. YAML header

Include:

- Title
- Your name as the *author*

2. Setup chunk

Load all required libraries (for example: `tidyverse`, `lubridate`, `ggplot2`, `nycOpenData`, etc.).

3. Minimum required code chunks (4)

Your report must contain *at least* the following four code chunks:

1. Data ingest

Use `nycOpenData::nyc_shooting_incidents()` to load the NYPD shootings dataset.

2. Data cleaning

At least **two** cleaning/feature steps (examples: dropping NAs, creating `time_of_day`, lowercasing a column, creating `days_since`).

3. Insights

Include the code used to generate your Week 3 insights (grouped summaries, counts, etc.).

4. Tables & Graphs

- At least **one table** created with `knitr::kable()`
- At least **two ggplot2 graphs**:

- one related to `time_of_day`
- one supporting your personal insight

All tables and graphs must have captions!

4. Narrative text

After **each** code chunk, write a short explanation describing:

- what you did
- why you did it
- what you learned or produced

5. Inline R expression

Include at least **one** inline R expression in your narrative text.

Example:

The dataset contains `r nrow(shooting_data)` rows after cleaning.

6. Hyperlink to the dataset

Provide a clickable link to the dataset (not a raw URL pasted as plain text).

Example:

[NYC Open Data Portal — NYPD Shooting Incident Data](#)

(You may link directly to the dataset page if you prefer.)

Reflection

In **3–5 sentences**, explain how this workflow could potentially help you in your thesis research.

Publishing Instructions

1. Render your Quarto document to HTML.
 2. Publish it online (e.g., RPubS).
 3. Submit:
 - the link to the published HTML
 - your `.qmd` file
-

Reproducibility Practice (Quarto Reporting)

This assignment focuses on reproducibility as **communication**.

In your report:

- Chunk labels should be meaningful (e.g., `data-ingest`, `cleaning`, `insights`, `plots`)
 - Your narrative should match your code (a reader should understand what each chunk accomplishes)
 - Inline R values should be used to avoid hard-coding numbers that may change if the dataset updates
 - Your HTML output should render cleanly and be readable without needing the original script
-

Publishing Instructions

1. Render your `.qmd` file to HTML.
2. Confirm that warnings and messages do not appear in the final document.
3. Submit both your `.qmd` file and your published `.html` report.

Assignment 6: ANOVA and Payment Patterns in NYC Camera Violations

Overview

You are working as a data scientist hired by a law firm that specializes in fighting parking and camera tickets.

The firm wants to understand whether certain groups tend to pay more than others. Previously, we explored patterns by day of week and time of day. Now, you will conduct a **structured ANOVA investigation** using NYC camera violation data.

Your job is to determine whether payment differences across groups are:

- Statistically significant
- Practically meaningful
- Strategically useful for marketing

Be sure to include a hyperlink somewhere in your introduction to the dataset via the [NYC Open Data Portal](#).

Learning Objectives

By completing this assignment, you will be able to:

- Load public NYC data using an R package
 - Clean and prepare real-world categorical and numeric variables
 - Engineer time-based features (day of week, time of day)
 - Conduct multiple one-way ANOVA tests
 - Interpret F-values, p-values, and PRE
 - Distinguish between statistical and practical significance
 - Communicate inferential findings clearly in a reproducible report
-

Textbook Connection

This assignment builds directly from **Chapter 5: Comparing Multiple Groups** in [Reproducible Research Using R](#).

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

Submission Instructions

Submit **two** files:

1. Your `.qmd` file
2. Your knitted `.html` file

Your document must:

- Knit without errors
 - Contain clearly labeled sections
 - Include visualizations, descriptive statistics, and ANOVA results
 - Include written interpretation after each major section
 - Use at least one inline R expression
-

Assignment Tasks

You must complete the following sections in order.

1. Data Ingest

Load the NYC camera violations dataset using the `nycOpenData` package.

Your section must:

- Clearly state what the dataset represents
- Report how many rows are in the dataset (use an inline R expression)
- Confirm that the dataset loaded successfully

Do not hard-code numbers in your text.

2. Data Cleaning and Feature Engineering

You must perform the following cleaning steps. You are responsible for determining how to implement them.

2A. Numeric Conversion

Ensure all payment-related variables (such as `payment_amount` and `fine_amount`) are stored as numeric.

Include a short checkpoint confirming the structure is correct.

2B. Date Cleaning

The `issue_date` column may contain inconsistent formats.

You must:

- Retain only properly formatted dates
- Convert `issue_date` into a Date object
- Create a new variable called `day_of_week`

Explain briefly why cleaning dates is necessary before modeling.

2C. Time Cleaning

Using the `violation_time` column:

- Convert it into a format that can be interpreted numerically
- Create a new categorical variable called `time_of_day`

Your categories must include:

- Morning
- Afternoon
- Night

Include a checkpoint showing how many observations fall into each category.

Explain why feature engineering improves interpretability.

3. Analysis 1: Day of Week and Fine Amount

You will now investigate whether average **fine__amount** differs by **day__of__week**.

Your section must include:

- A grouped descriptive summary (means displayed in descending order)
- A one-way ANOVA
- A `supernova()` table
- A post-hoc test (Tukey)

Then write a short paragraph answering:

- Is the effect statistically significant?
 - How much variance is explained (PRE)?
 - Is the effect practically meaningful?
 - Why might statistical significance be misleading here?
-

4. Analysis 2: Time of Day and Fine Amount

Repeat the same structure, but now examine **fine__amount** by **time__of__day**.

Include:

- Visualization (boxplot required)
- Descriptive statistics
- ANOVA
- PRE interpretation

In your paragraph, explain:

- Whether differences are meaningful

- Why fine amounts may show significance even if policy does not change
-

5. Analysis 3: Violation Type and Fine Amount

Examine **fine_amount** by violation type.

Your section must include:

- Descriptive statistics
- ANOVA
- PRE interpretation

In your paragraph, explain:

- Why this model is statistically strong
 - Why it may not be conceptually interesting
 - What this teaches us about modeling fixed policy variables
-

6. Analysis 4: Violation Type and Payment Amount

Now shift to the more behaviorally meaningful variable: **payment_amount**.

Examine **payment_amount** by violation type.

Your section must include:

- Visualization (boxplot required)
- Descriptive statistics
- ANOVA
- PRE interpretation

In your paragraph, explain:

- Why this analysis is more practically meaningful than `fine_amount`
 - What the size of PRE suggests
 - How a law firm might use this insight strategically
-

Final Reflection

In 5–7 sentences, answer:

1. What is the difference between statistical significance and practical significance?
 2. Which of the four analyses above do you think is most useful for real-world decision-making?
 3. If you were advising the law firm, what is one concrete recommendation you would make based on your findings?
-

Reproducibility Practice

This assignment focuses on reproducible statistical modeling.

Your document must:

- Use clearly labeled code chunks
- Avoid hard-coding values in your narrative
- Use inline R expressions where appropriate
- Include at least one data-checking “checkpoint” in each major section
- Render cleanly to HTML without manual intervention

The goal is that another analyst could rerun your entire workflow and verify every statistical conclusion.

Publishing Instructions

1. Render your `.qmd` file to HTML.
2. Confirm that warnings and messages do not appear in the final document.
3. Submit both your `.qmd` file and your published `.html` report.

Midterm: Exercise & Sleep

Overview

You are a researcher in sleep expert **Matthew Walker's lab**. You have just run an experiment measuring the effect of different **exercise methods** on sleep habits.

You measured:

- Average hours of sleep **before** the experiment
- Average hours of sleep **after** the experiment
- **Sleep efficiency** at the end of the experiment

Dr. Walker needs you to identify **which (if any) exercise method best improves sleep**, and to make a clear, evidence-based recommendation.

This midterm assesses your ability to build a **fully reproducible analysis** from raw data through statistical inference and final reporting.

Learning Objectives

By completing this midterm, you will be able to:

- Import data from a multi-sheet Excel workbook
 - Clean messy labels and standardize categorical variables
 - Merge datasets into a single analysis-ready table
 - Engineer derived variables and handle missing data responsibly
 - Produce descriptive statistics and professional tables
 - Create clear, interpretable visualizations
 - Conduct and interpret t-tests and ANOVAs
 - Run and interpret Tukey post-hoc comparisons
 - Synthesize results into an actionable recommendation
-

Textbook Connection

This assignment builds directly from **Chapters 1-5 (and 10)** in [Reproducible Research Using R](#).

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

Submission Instructions

Submit **two** files:

1. Your `.qmd` file
2. Your knitted and published `.html` file

Your report must:

- Knit/render without errors
 - Include short narrative text between code chunks
 - Use clear section headers matching the numbered tasks below
 - Suppress unnecessary warnings/messages in the final HTML
 - Include readable plots and (recommended) formatted tables (e.g., `kable()`)
-

Midterm Tasks

1. Setup & Data Import

Tasks

- Load all packages you use
 - Import **both** sheets from `midterm_sleep_exercise.xlsx` using `readxl`:
 - `participant_info_midterm`
 - `sleep_data_midterm`
 - Preview both datasets using appropriate functions (e.g., `glimpse()` or `head()`)
-

2. Merge & Base Cleaning

Tasks

- Clean column names so they are consistent and easy to work with
- Standardize labels for:
 - `Exercise_Group` (e.g., “cardio”, “CARDIO” → “Cardio”)
 - `Sex` (e.g., “m”, “MALE” → “Male”)
- Merge the two datasets into a single dataset

Requirements

- Final dataset should be **one row per participant**
 - Dataset should include demographics + sleep variables
 - Include a checkpoint verifying row count and that the merge behaved as expected
-

3. Create Derived Variables

Tasks

- Create a numeric sleep-change variable:
 - `Sleep_Difference = post_sleep - pre_sleep_num`
- Create `AgeGroup2` using `case_when()` with exactly **two groups** (example: `< 40` vs `>= 40`)
- Identify how many rows are missing `Sleep_Difference`
- Drop rows with missing `Sleep_Difference`

Requirements

- Include a checkpoint showing how many rows were removed and why
 - Your narrative should briefly justify your missing-data decision
-

4. Descriptive Statistics

Tasks

Report descriptive statistics for:

- `Sleep_Difference` (overall)
- `Sleep_Efficiency` (overall)

For each variable, include:

- Mean
- SD
- Min
- Max

Then report group-wise summaries by `Exercise_Group` for:

- `Sleep_Difference`
- `Sleep_Efficiency`

Requirements

- Group summaries must be clearly readable
- Tables should be formatted professionally using `kable()` and appropriate captions.

5. Visualizations (3 plots)

Create **three** plots:

1. Boxplot: `Sleep_Difference` by `Exercise_Group`
2. Boxplot: `Sleep_Efficiency` by `Exercise_Group`
3. Scatterplot: `Sleep_Difference` vs `Sleep_Efficiency`, with a trend line

Requirements

Each plot must include:

- Descriptive title
- Clear x- and y-axis labels
- A clean theme (e.g., `theme_minimal()`)
- Appropriate captions

After each plot, write 1–3 sentences describing what the plot suggests.

6. t-tests (Two)

Run **two** t-tests:

1. `Sleep_Difference ~ Sex`
2. `Sleep_Difference ~ AgeGroup2`

Requirements

For each t-test, report:

- Group means
- t statistic
- df
- p-value

Then interpret in plain language:

- Is the difference statistically significant?
 - Is it practically meaningful?
-

7. ANOVAs (Two) + Post-hoc Tests

Run **two** ANOVAs:

- ANOVA A: `Sleep_Difference ~ Exercise_Group`
- ANOVA B: `Sleep_Efficiency ~ Exercise_Group`

Requirements

For each ANOVA, include:

- ANOVA table output
- F statistic
- df
- p-value
- A brief effect-size comment
- PRE from `supernova()` (required)

Then run Tukey post-hoc tests:

- `TukeyHSD()` for each ANOVA

Interpret results:

- Which groups differ significantly?
 - Which exercise group appears to “win” for each outcome?
-

8. Synthesis & Recommendation

Write **4–6 sentences** answering:

- Considering **both** outcomes (`Sleep_Difference` and `Sleep_Efficiency`) and your post-hoc results, which **single** exercise regimen would you recommend to improve overall sleep?

Requirements

- You may only pick **one** regimen
- Support your recommendation with specific statistical evidence:
 - F values and p-values
 - Key Tukey contrasts
 - Any meaningful patterns from your plots/tables

Your recommendation must be actionable and executive-friendly.

9. Reflection

Write **3–5 sentences** addressing:

- What was most challenging?
 - What did you feel confident about?
 - What would you do differently next time to improve your analysis or report?
-

Reproducibility Practice

This midterm evaluates not only statistical accuracy, but workflow transparency. Your report must:

- Load all packages explicitly
 - Import both sheets programmatically
 - Confirm merge success with row counts
 - Explicitly handle missing values (no silent removal)
 - Use clearly labeled code chunks
 - Avoid hard-coding statistical values in written interpretation
 - Suppress unnecessary warnings/messages
 - Render cleanly to HTML
-

Publishing Instructions

1. Render your `.qmd` file to HTML.
2. Confirm warnings and messages do not appear in the final document.
3. Submit both your `.qmd` file and your published `.html` report.

Assignment 7: NBA Analytics — Exploring Team Performance Through Reproducible Analysis

Overview

The NBA season is about to begin, and you have just been hired as a **Data Analyst for the National Basketball Association**. Commissioner **Adam Silver** wants insights into last season's team performances — specifically:

- How offensive and defensive metrics relate to one another
- Whether teams from the Eastern and Western Conferences differ
- Whether team-level characteristics help explain performance patterns

You have been given an Excel workbook containing all 30 NBA teams, with each team stored on a separate sheet. Your job is to build a **fully reproducible Quarto analysis** that loads, cleans, visualizes, and analyzes the data.

Your final product should read like a professional analytics report prepared for an executive audience.

Learning Objectives

By completing this assignment, you will be able to:

- Write reusable functions in R for structured data ingestion
 - Load multi-sheet Excel workbooks programmatically
 - Engineer new performance metrics
 - Merge datasets using lookup tables
 - Conduct correlation and partial correlation analyses
 - Interpret statistical relationships in plain language
 - Communicate findings clearly to a non-technical executive audience
 - Produce a fully reproducible Quarto document
-

Textbook Connection

This assignment builds directly from **Chapter 7: Correlation** in [Reproducible Research Using R](#).

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

Submission Instructions

Submit **two** files:

1. Your `.qmd` file
2. Your knitted `.html` file

Your document must:

- Knit without errors
 - Contain clearly labeled sections
 - Include all required visualizations
 - Include all correlation analyses
 - Include a professional memo to Adam Silver
 - Use at least one inline R expression
-

Assignment Tasks

1. Data Ingest and Preparation

You must programmatically load all 30 teams from the Excel workbook.

Required Tasks

Create a function that:

- Loads a single sheet from the workbook
- Adds a new column identifying the **Team** (using the sheet name)
- Creates a binary column called **Won_award** (0 = No, 1 = Yes)
- Calculates the following metrics:
 - **PRA** = **Points** + **Rebounds** + **Assists** (offensive metric)
 - **STOCKS** = **Steals** + **Blocks** (defensive metric)

Then:

- Use `excel_sheets()` to list all sheet names
- Apply your function to every sheet using `lapply()`
- Combine all teams into one complete dataset using a row-binding method

Requirements

- Your code must not manually load sheets one-by-one
- The final dataset must include all 30 teams
- Include a checkpoint confirming the total number of rows
- Report the total number of observations using an inline R expression

Briefly explain why writing a function improves reproducibility.

2. Adding Conference Information

You are provided with a Conference lookup sheet.

You must:

- Merge Conference information into your combined dataset
- Recode Conference as a binary variable for analysis:
 - East = 1
 - West = 0

Include a small checkpoint confirming that both conferences are represented correctly.

Briefly explain why binary coding is useful for correlation analysis.

3. Visual Exploration

Create **at least two visualizations** using `ggplot2`.

Required Plot 1

- A scatterplot showing the relationship between PRA and STOCKS
- Points must be colored by Conference

Required Plot 2

- A second visualization of your choice
(Examples: distribution of PRA by conference, age vs PRA, STOCKS vs Age, etc.)

For each plot, write 3–4 sentences describing:

- What patterns you observe
- Whether offensive and defensive metrics appear related
- Whether conferences appear visually different

All plots must include:

- Clear titles
 - Axis labels
 - A caption
-

4. Correlation Analysis

A. Point-Biserial Correlation

Test whether Conference (East = 1, West = 0) is related to:

- PRA
- STOCKS

Report:

- Correlation coefficient (r)
 - Direction (positive or negative)
 - Strength (weak, moderate, strong)
 - Statistical significance
 - Plain-language interpretation
-

B. Correlation Matrix

Create a small correlation matrix including at least:

- Age
- PRA
- STOCKS

Visualize the matrix using `ggcorrplot`.

Then answer:

- Which relationships are strongest?
 - Are offensive and defensive metrics strongly related?
 - Does Age appear meaningfully associated with performance?
-

C. Partial Correlation

Run a partial correlation examining the relationship between:

- PRA and STOCKS
- While controlling for either Age or Minutes Played

Interpret:

- How the relationship changes when controlling for the third variable
- Whether the association appears independent of that control variable

Explain in plain language what “controlling for” means.

5. Executive Memo to Adam Silver

Write a short professional memo addressed to:

Adam Silver
Commissioner, National Basketball Association

Your memo must include:

- A summary of your key findings
- Whether Eastern and Western Conference teams appear statistically different
- Whether offensive and defensive performances tend to move together
- One limitation of your analysis
- One recommended next step

Your tone should be professional and executive-level, not academic.

Guidelines for Interpretation

When interpreting correlations, always include:

- The direction (positive or negative)
- The strength (weak, moderate, strong)
- Whether it is statistically significant
- A plain-language explanation

Example of plain-language style:

As PRA increases, STOCKS also tends to increase, suggesting that teams with stronger offensive production may also contribute more defensive playmaking.

Reflection

In 4–6 sentences, answer:

1. Why is reproducible workflow especially important in sports analytics?
 2. How does programmatically loading data (rather than manual steps) improve reliability?
 3. What did you learn about interpreting correlations versus visual patterns?
-

Reproducibility Practice

This assignment emphasizes reproducible data pipelines.

Your document must:

- Load all sheets programmatically
- Avoid manual data entry
- Use clearly labeled code chunks
- Avoid hard-coding values in written interpretation
- Use inline R expressions where appropriate
- Render cleanly to HTML

The goal is that another analyst could rerun your entire workflow and verify your findings without guessing what steps were taken.

Publishing Instructions

1. Render your `.qmd` file to HTML.
2. Confirm that warnings and messages do not appear in the final document.
3. Submit both your `.qmd` file and your published `.html` report.

Assignment 8: Florida Crime Analytics — Uncovering the Root of Florida's Crime Surge

Overview

The **Florida Police Department (FPD)** has hired you as their new **Data Analyst**. Your mission is to uncover which socioeconomic factors are most strongly associated with rising crime rates across Florida counties.

Using county-level data, you will conduct a **fully reproducible Quarto analysis** to determine whether **income, education, or urbanization** best explain differences in crime rates.

Your findings will help inform statewide prevention strategies, resource allocation, and community outreach efforts.

This assignment assesses your ability to:

- Clean and prepare structured data
 - Compute and interpret correlations
 - Build and compare regression models
 - Evaluate model fit using R^2 and AIC
 - Communicate statistical findings to a non-technical executive audience
-

Learning Objectives

By completing this assignment, you will be able to:

- Clean and standardize real-world datasets
 - Perform exploratory data analysis (EDA)
 - Interpret correlation matrices
 - Build simple and multiple regression models
 - Compare models using R^2 , Adjusted R^2 , and AIC
 - Explain regression results in plain language
 - Produce a reproducible Quarto report suitable for executive review
-

Textbook Connection

This assignment builds directly from **Chapter 8: Linear Regression** in [Reproducible Research Using R](#).

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

Textbook Connection

This assignment builds directly from **Chapter 8: Linear Regression** in [Reproducible Research Using R](#).

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

Submission Instructions

Submit **two** files:

1. Your `.qmd` file
2. Your knitted and published `.html` file

Your document must:

- Knit without errors
 - Suppress warnings and messages in the final HTML
 - Include clearly labeled sections
 - Include all required visualizations
 - Include regression tables and model comparisons
 - Include a professional memo to the Chief of the FPD
 - Use at least one inline R expression
-

Assignment Tasks

1. Data Loading and Preparation

You have been provided with an Excel file containing the following columns:

Column	Description
County	County name
C	Crime rate (per 1,000 residents)
I	Median income (in thousands)
HS	High-school graduation rate (%)
U	Urban population (%)

Required Tasks

You must:

- Load the dataset reproducibly
- Rename the columns to:
 - Crime
 - Income
 - HighSchoolGrad
 - UrbanPop
- Format all county names so that only the first letter is capitalized
- Inspect the dataset structure
- Provide summary statistics

Include a brief explanation of why consistent naming and formatting are essential for reproducible modeling.

2. Exploratory Data Analysis

Conduct exploratory analysis to understand the structure and distribution of the data.

Required Components

- Compute descriptive statistics (mean, median, range) for all numeric variables
- Create at least **two visualizations** (e.g., histograms, boxplots, scatterplots)
- Provide written interpretation of what you observe

Your interpretation should address:

- Distribution shapes
- Potential outliers
- Initial patterns between predictors and crime

Bonus (Optional)

Using `ggplot2` and mapping tools, create a heatmap of Florida counties based on crime rates.

3. Correlation Analysis

Investigate which factors are most strongly associated with crime.

Required Tasks

- Compute a correlation matrix including:
 - Crime
 - Income
 - HighSchoolGrad
 - UrbanPop
- Identify which predictor shows the strongest relationship with Crime

- Interpret the direction (positive or negative)
- Interpret the strength (weak, moderate, strong)

(Optional) Visualize the correlation matrix using a correlation plotting tool.

In your written interpretation, explain:

- Which factor appears most important
 - Whether any predictors appear highly correlated with one another
 - Why correlation does not imply causation
-

4. Building Regression Models

You will now build models to predict county-level crime rates.

Required Models

1. A **simple regression model** using the predictor you believe fits best
2. At least **two multiple regression models**, such as:
 - Income + HighSchoolGrad
 - Income + UrbanPop
 - A full model with all predictors
3. (Optional) Include an interaction term if theoretically justified

Model Comparison

Compare models using:

- R^2
- Adjusted R^2
- AIC

Interpret:

- Which predictors are statistically significant

- Which model explains the most variance
- Which model best balances simplicity and explanatory power

Explain why Adjusted R^2 and AIC are useful when comparing multiple models.

5. Executive Memo to the Chief of the Florida Police Department

Write a short professional memo addressed to:

Chief
Florida Police Department

Your memo must include:

- The best-performing model
- The most influential predictor(s)
- The proportion of variance explained (R^2)
- Clear, plain-language interpretation
- One limitation of your analysis
- One actionable recommendation for resource allocation or prevention strategy

Your tone should be executive-level and professional.

Guidelines for Interpretation

When discussing results, always include:

- **Direction** — Positive or negative relationship
- **Strength** — Weak, moderate, or strong
- **Significance** — Statistically significant or not
- **Plain language meaning**

Example:

Counties with lower median income tend to have higher crime rates. For every \$1,000 increase in median income, predicted crime decreases while holding other factors constant.

Avoid technical jargon without explanation.

Final Question

At the end of your report, include a short written response to:

Based on your analysis, which model best predicts Florida's county-level crime rates, and why?

Be specific and reference your model comparison metrics.

Reflection

In 4–6 sentences, answer:

1. Why is reproducibility especially important when working with public policy data?
2. What did you learn about the difference between correlation and regression?
3. How does model comparison improve decision-making?

Reproducibility Practice

This assignment emphasizes reproducible modeling and executive communication.

Your document must:

- Load data programmatically
- Avoid manual data manipulation outside the script
- Use clearly labeled code chunks
- Avoid hard-coding results in narrative
- Use inline R expressions where appropriate
- Suppress warnings/messages in the final published HTML

The goal is that another analyst could rerun your analysis and verify every claim.

Publishing Instructions

1. Render your `.qmd` file to HTML.
2. Confirm that warnings and messages do not appear in the final document.
3. Submit both your `.qmd` file and your published `.html` report.

Assignment 9: Streaming Analytics — Understanding Platform Popularity Across Age Groups

Overview

The **Streaming Analytics Division (SAD)** has hired you as their new **Data Analyst**. Your mission is to determine whether **age group influences streaming platform preference**.

The company wants to know whether certain platforms — Netflix, Hulu, Disney+, Amazon, or Other — appeal more strongly to specific age demographics (18–25, 26–40, 41+). Your analysis will guide targeted marketing strategies, promotional campaigns, and content investment decisions.

Using simulated survey data, you will conduct a **fully reproducible Quarto analysis** to determine whether **Age Category and Platform Preference** are statistically related.

You will:

- Construct and interpret contingency tables
 - Perform a Chi-Square test of independence
 - Examine observed, expected, and residual values
 - Decompose the Chi-Square statistic into contributions
 - Visualize contributions using a heatmap
 - Compute and interpret Cramer's V
 - Communicate findings clearly in plain language
-

Learning Objectives

By completing this assignment, you will be able to:

- Clean and summarize categorical data
 - Construct contingency tables
 - Conduct and interpret a Chi-Square test of independence
 - Interpret observed, expected, and residual frequencies
 - Decompose the Chi-Square statistic into cell-level contributions
 - Visualize categorical data using stacked and clustered bar charts
 - Calculate and interpret Cramer's V
 - Communicate statistical findings in accessible language
-

Textbook Connection

This assignment builds directly from **Chapter 6: Analyzing Categorical Data** in [Reproducible Research Using R](#).

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

Submission Instructions

Submit **two files**:

1. Your `.qmd` file
2. Your knitted and published `.html` file

Your document must:

- Knit without errors
 - Suppress warnings and messages in the final HTML
 - Include clearly labeled sections
 - Include all required tables and visualizations
 - Include full statistical interpretations
 - Use at least one inline R expression
-

Assignment Tasks

1. Data Preparation

You have been provided with survey data including:

- Age Category (18–25, 26–40, 41+)
- Preferred Streaming Platform (Netflix, Hulu, Disney+, Amazon, Other)

Required Tasks

You must:

- Inspect the dataset
- Compute total counts for Age Category
- Compute total counts for Platform Preference
- Construct a contingency table showing the joint distribution of Age Category and Platform Preference

Briefly describe what the contingency table represents and why it is useful for categorical analysis.

2. Visualization

Create **two visualizations** using `ggplot2`.

Required Visualization 1

A **stacked bar chart** showing the proportions of platform preferences within each age group.

Required Visualization 2

A **clustered (side-by-side) bar chart** showing counts for each platform across age groups.

Both visualizations must:

- Include descriptive titles
- Include labeled axes
- Include a legend
- Use a professional or themed style (e.g., from `ggthemes`)

After each plot, provide 3–4 sentences interpreting what you observe.

3. Chi-Square Test of Independence

Conduct a **Chi-Square test of independence** to determine whether Age Category and Platform Preference are related.

In your write-up, report and interpret:

- The Chi-Square statistic (χ^2)
- Degrees of freedom (df)
- The p-value

Then answer:

- Is there a statistically significant relationship?
 - What does this result mean in plain language?
-

4. Observed, Expected, and Residual Values

From your Chi-Square test, extract:

- Observed counts
- Expected counts
- Residuals

Discuss:

- Which age–platform combinations show more people than expected
- Which show fewer people than expected
- Which residuals appear largest in magnitude

Explain what residuals tell us about patterns in the data.

5. Contributions to the Chi-Square Statistic

Break down the Chi-Square statistic into **individual cell contributions**.

Then:

- Calculate the percentage contribution of each cell to the total χ^2 value
- Identify which age–platform combinations contribute most to the overall result

Visualize these percentage contributions using a **heatmap**.

In your interpretation, explain:

- Which cells drive the overall Chi-Square result
 - What this suggests about viewing habits across age groups
-

6. Effect Size (Cramer's V)

Compute **Cramer's V** to measure the strength of association between Age Category and Platform Preference.

Interpret the value using these guidelines:

- 0.00–0.10 → Very Weak
- 0.10–0.30 → Weak
- 0.30–0.50 → Moderate
- Above 0.50 → Strong

Provide a short, plain-language explanation of the effect size.

7. Final Interpretation

Write a professional summary addressing:

- Whether the relationship between Age and Platform is statistically significant
- Which age groups and platforms are driving the relationship
- The strength of the association (Cramer's V)
- What these findings mean in a real-world marketing context

Your summary should resemble a professional analytics report, not a textbook answer.

Deliverables

Your final report must include:

- The contingency table
 - Both visualizations (stacked and clustered)
 - Chi-Square results (χ^2 , df, p-value)
 - Observed, expected, and residual tables
 - Contribution table and heatmap
 - Cramer's V calculation and interpretation
 - Final written summary
-

Guidelines for Interpretation

When discussing your results, always include:

- **Statistical significance** — Is the relationship real?
- **Direction of pattern** — Which combinations are over- or under-represented?
- **Strength of association** — Using Cramer's V
- **Plain-language meaning** — What does this imply about streaming habits?

Avoid jargon without explanation.

Reflection

In 4–6 sentences, answer:

1. Why is the Chi-Square test appropriate for categorical data?
2. What is the difference between statistical significance and effect size?
3. Why is it important to examine residuals and contributions rather than only the p-value?

Reproducibility Practice

This assignment emphasizes transparent categorical analysis.

Your document must:

- Construct tables programmatically
- Avoid manual calculations outside your script
- Use clearly labeled code chunks
- Avoid hard-coding statistical results in narrative
- Use inline R expressions where appropriate
- Render cleanly to HTML

The goal is that another analyst could reproduce your entire Chi-Square analysis and verify every conclusion.

Publishing Instructions

1. Render your `.qmd` file to HTML.
2. Confirm warnings and messages do not appear in the final document.
3. Submit both your `.qmd` file and your published `.html` report.

Assignment 10: Wage Analytics — Predicting High vs Low Earners Using Logistic Regression

Overview

The **Wage Analytics Division (WAD)** has hired you as their new **Data Scientist**.

Your mission is to determine whether demographic and employment characteristics can be used to **predict whether a worker earns a high or low wage**.

Using the **Wage** dataset from the ISLR package, you will:

- Create a binary wage classification
- Explore relationships between predictors and wage
- Build a logistic regression model
- Evaluate predictive performance on unseen data

Your analysis will help the company understand which characteristics are associated with higher earnings and determine how well a predictive model performs in practice.

Learning Objectives

By completing this assignment, you will be able to:

- Clean and prepare mixed-type real-world data
 - Create a binary outcome variable
 - Conduct t-tests, ANOVA, and Chi-Square tests
 - Perform a train/test split
 - Build and interpret a logistic regression model
 - Evaluate classifier performance using confusion matrices and ROC/AUC
 - Communicate predictive modeling results clearly and professionally
-

Textbook Connection

This assignment builds directly from **Chapter 9: Logistic Regression** in [Reproducible Research Using R](#).

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

Submission Instructions

Submit **two files**:

1. Your `.qmd` file
2. Your knitted and published `.html` file

Your document must:

- Knit without errors
 - Suppress warnings and messages in the final HTML
 - Include clearly labeled sections
 - Include all statistical tests and model outputs
 - Include performance metrics and ROC curve
 - Include a professional final summary
 - Use at least one inline R expression
-

Assignment Tasks

1. Create the WageCategory Variable

Begin by preparing the dataset.

Required Tasks

- Load the `Wage` dataset from the ISLR package
- Create a new factor variable called `WageCategory`:
 - If `wage > median(wage)` → "High"
 - If `wage < median(wage)` → "Low"
- Convert `WageCategory` into a factor variable

Important

- You must **NOT** remove `wage` or `logwage`
- Be prepared to explain the concept of **target leakage** and why it matters in predictive modeling

Briefly explain why creating a binary outcome transforms this into a classification problem.

2. Data Cleaning

Some categorical variables in the `Wage` dataset contain numeric prefixes (e.g., "3. Asian").

Required Tasks

- Remove numeric prefixes from all relevant categorical variables
- Ensure categorical variables are clean and properly formatted before analysis

Provide a brief explanation of why cleaning categorical labels is important for interpretation and modeling.

3. Classical Statistical Tests

You will explore relationships between predictors and wage using classical inference methods.

A. t-test

Compare **age** between High and Low wage earners.

Your write-up must include:

- Mean age for each wage group
 - t-statistic, degrees of freedom, and p-value
 - A short interpretation of whether age differs between wage categories
-

B. ANOVA

Using the original numeric **wage** variable, choose **one** of the following:

- `wage ~ education`
- `wage ~ maritl`
- `wage ~ jobclass`

Your write-up must include:

- ANOVA table
- F-statistic, degrees of freedom, and p-value
- A brief interpretation of the result

C. Chi-Square Test

Choose **one categorical relationship** that is different from your ANOVA variable.

You must:

- Create a contingency table
- Conduct a Chi-Square test
- Report χ^2 , df, and p-value
- Compute and interpret Cramer's V
- Describe any patterns observed in the contingency table

Explain what the Chi-Square test tells you about association.

4. Logistic Regression Model

You will now build a predictive model.

Train/Test Split

- Perform a 70/30 split using `sample.split()`
- Clearly define your training and testing datasets

Explain why separating training and test data is critical in predictive modeling.

Model Specification

- Build a logistic regression model predicting `WageCategory`
- Choose predictors thoughtfully (be cautious of leakage or redundancy)

Your write-up must include:

- Logistic regression output
 - Odds ratios using `exp(coef())`
 - A 2–4 sentence interpretation describing:
 - Which predictors are significant
 - Direction of effects
 - Magnitude of effects
 - Any surprising findings
-

5. Model Evaluation on Test Data

Using your **test dataset**, compute:

- Predicted probabilities
- Predicted classes (“High” or “Low”)
- Confusion matrix

Report and interpret:

- Accuracy
 - Sensitivity
 - Specificity
 - Balanced Accuracy
-

ROC Curve and AUC

- Generate an ROC curve
- Compute AUC

In your interpretation, discuss:

- How well the model identifies high earners
 - How accuracy compares to the No Information Rate
 - Whether the model performs better than guessing
-

6. Final Interpretation

Write a professional 3–5 paragraph summary addressing:

- Whether age, education, jobclass, or marital status meaningfully relate to wage
- Which predictors were most influential in the logistic regression
- How well the model performed on unseen data
- Which wage group the model predicts best
- What variables you would add or remove if repeating the analysis
- One limitation of your modeling approach

Your tone should resemble a professional data science report, not a homework answer.

Guidelines for Interpretation

When interpreting regression results, include:

- Direction (positive or negative relationship)
- Magnitude (odds ratio interpretation)
- Statistical significance
- Plain-language meaning

Example:

Workers with higher education levels have greater odds of being classified as high earners, holding other factors constant.

Avoid technical jargon without explanation.

Reflection

In 4–6 sentences, answer:

1. What is the difference between inference and prediction?
2. Why is target leakage dangerous in predictive modeling?
3. What did you learn about evaluating classifier performance?

Reproducibility Practice

This assignment emphasizes responsible predictive modeling.

Your document must:

- Load data programmatically
- Clean variables reproducibly
- Use clearly labeled code chunks
- Avoid hard-coding values in narrative
- Use inline R expressions where appropriate
- Suppress warnings and messages in the final HTML
- Render cleanly without manual edits

The goal is that another analyst could reproduce your entire workflow and validate every statistical claim.

Publishing Instructions

1. Render your `.qmd` file to HTML.
2. Confirm warnings and messages do not appear in the final document.
3. Submit both your `.qmd` file and your published `.html` report.

Final Project: NYC Open Data 2026 Proposal

Overview

Your final project for this course is to design and conduct an **original research project** using NYC Open Data.

Your project will be written as a fully reproducible **Quarto document** and structured as though it were ready for public presentation at **NYC Open Data Week**.

This assignment is designed to:

- Give you experience designing an independent research project
- Provide hands-on work with open civic datasets
- Produce a polished portfolio-ready artifact
- Strengthen your reproducible workflow and data communication skills
- Prepare you to submit to NYC Open Data Week 2026 if you choose

By the end of this course, you will have created a professional-quality research document suitable for public presentation.

Learning Objectives

By completing this project, you will be able to:

- Formulate a clear, data-driven research question
 - Identify and justify appropriate open datasets
 - Build a fully reproducible data workflow
 - Clean, wrangle, and analyze civic data responsibly
 - Create clear and meaningful visualizations
 - Write for a public-facing, non-technical audience
 - Connect your work to broader open data principles
-

Textbook Connection

This final project synthesizes concepts in [Reproducible Research Using R](#).

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

Project Requirements

1. Dataset Selection

You must:

- Select **at least one dataset** from the NYC Open Data Portal (preferred).
- Other publicly available civic datasets regarding NYC are acceptable **with instructor approval**.
- Clearly cite and hyperlink all datasets used.
- Justify why the dataset is appropriate for your research question.

Your project should demonstrate thoughtful dataset selection rather than convenience-based selection.

2. Reproducible Analysis (Quarto)

Your final submission must:

- Be written in **Quarto (.qmd)** format.
- Include narrative, code, output, and interpretation in one document.
- Knit/render successfully without errors.
- Suppress unnecessary warnings and messages in the final output.

Your workflow should clearly demonstrate:

- Data import
- Cleaning and transformation

- Analysis aligned with your research question
 - Transparent decision-making
-

3. Proposal-Style Structure

Your document should be structured as if it were an Open Data Week proposal.

Include the following sections:

A. Title & Event Description

- A clear, compelling project title.
- A 1–2 paragraph overview explaining:
 - Your research question
 - Why it matters
 - What attendees would learn

Write this as though it were submitted to Open Data Week.

B. Dataset(s) Used

- Describe the dataset(s).
 - Provide proper citations and links.
 - Explain how the dataset(s) connect to your research question.
-

C. Analysis

- Present a reproducible workflow from raw data to results.
 - Clearly explain:
 - What you cleaned and why
 - What transformations were performed
 - What analyses were conducted
 - Align every analysis step with your research question.
-

D. Visualizations

Include **at least two** clear, well-labeled figures or tables created in R.

All visualizations must:

- Include titles
- Include axis labels
- Include captions
- Be readable in a public-facing setting

Explain what each visualization shows and why it matters.

E. Audience & Relevance

Identify:

- Who would care about this project?
 - New Yorkers?
 - Policymakers?
 - Journalists?
 - Community organizations?
 - Why does this research matter in a civic context?
-

F. Connection to Open Data

Explain how your project:

- Demonstrates transparency
 - Highlights accessibility of public data
 - Shows the value of open civic information
 - Reflects open data principles
-

Length & Formatting Requirements

- Minimum **500 words of written narrative** (excluding code).
- Figures and tables must be clearly labeled.
- Citations must be properly formatted.
- The document must render cleanly to HTML or PDF.

Submit both:

- Your `.qmd` file
 - Your knitted HTML or PDF output
-

Class Presentation

You will present your project during the final weeks of the semester.

Presentation Requirements:

- 10–12 minutes
- 2–3 minutes for Q&A
- Use slides (Quarto, PowerPoint, etc.)

Your presentation should:

- Clearly explain your research question
- Highlight key findings
- Show at least one visualization
- Be understandable to a non-technical audience

This is practice for a potential Open Data Week 2026 submission.

Class Publication (Optional but Encouraged)

At the end of the semester, projects may be compiled into a collective class book via Posit Cloud.

Each student's project may appear as a chapter.

Benefits:

- A publication credit
- A portfolio artifact
- A public-facing research contribution

You may opt out if you prefer.

Grading (100 Points Total)

- Dataset & Research Question (15 pts)
Clarity, appropriateness, creativity
 - Reproducibility (20 pts)
Working document, organized workflow, transparent code
 - Analysis (20 pts)
Depth, correctness, alignment with question
 - Visualizations (10 pts)
Clarity, design, relevance
 - Public Framing & Civic Relevance (10 pts)
Event description, audience, connection to Open Data
 - Class Presentation (15 pts)
Clarity, engagement, professionalism
 - Professionalism & Formatting (10 pts)
Writing quality, citations, organization
-

Reproducibility Practice

This final project emphasizes transparent and repeatable workflows.

Your document must:

- Load all required packages explicitly
- Import data programmatically
- Avoid manual editing of datasets
- Clearly document cleaning decisions
- Avoid hard-coding results in interpretation
- Render cleanly without errors
- Set a seed if using sampling, modeling, or randomness

The goal is that another analyst could reproduce your findings exactly.

Resources

- NYC Open Data Portal: <https://opendata.cityofnewyork.us/>
 - Past Open Data Week Events: <https://2025.open-data.nyc/>
-

Submission

Required:

- .qmd file
 - Knitted HTML or PDF output
-

Takeaway

By completing this final project, you will leave the course with:

- An independent, reproducible research project
- A polished presentation
- A potential publication credit
- A strong foundation for an Open Data Week 2026 submission

Final Assignment: Building Your Reproducible Quarto Book

Overview

Your final assignment is to transform all of your completed course work into a polished, cohesive **Quarto book**.

Throughout the semester, every assignment was written in Quarto format for a reason. Now, you will combine them into a single published book.

You will:

- Convert your assignments into a structured Quarto book project
- Organize them into chapters
- Polish formatting and presentation
- Publish your completed book

This book will serve as a **professional portfolio artifact** that you can share with:

- Potential employers
- Graduate programs
- Research labs
- Collaborators
- Friends and family

By completing this assignment, you are not just submitting coursework —

You are officially becoming an author.

Learning Objectives

By completing this project, you will:

- Organize multiple Quarto documents into a unified book structure
 - Apply consistent formatting and narrative across chapters
 - Improve clarity, presentation, and documentation
 - Practice professional-level data communication
 - Publish a complete analytical portfolio
-

Textbook Connection

This final assignment synthesizes concepts in [Reproducible Research Using R](#).

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

Assignment Requirements

1. Create a Quarto Book Project

You must:

- Create a Quarto book project (`_quarto.yml` configured as a book)
- Organize your assignments as individual chapters
- Ensure the book renders successfully from start to finish
- Publish your completed book (via your chosen publishing method)

Your book must render cleanly without errors.

2. Required Chapters (In This Order)

Your book must include the following chapters, in this exact order:

1. R Script → R Markdown Report
2. Homework: Continuing the Law Firm Analysis (Part 2)
3. Midterm
4. NBA Analytics – Exploring Team Performance Through Reproducible Analysis
5. Florida Crime Analytics – Uncovering the Root of Florida’s Crime Surge
6. Streaming Analytics – Understanding Platform Popularity Across Age Groups
7. Wage Analytics – Predicting High vs Low Earners Using Logistic Regression

8. RMarkdown, Shiny, and NYC 311 Data

Each chapter must be its own `.qmd` file.

3. Title Page

Your book must include a professional title page containing:

- A clear, descriptive book title
- Your full name
- Course name and semester
- A short “**About This Book**” section (2–4 sentences)

This page should feel polished and intentional.

4. Cover Image

Your book must include a cover image.

Your cover image can be:

- Something you designed
- An image related to your analyses
- A professional photo or graphic
- A visual that represents your identity as a data analyst

The image must be appropriate and professional.

5. Chapter Introductions

Each chapter must begin with a short introduction explaining:

- The purpose of the assignment
- The skills or methods used
- Any relevant context for the reader

If the original assignment does not contain a chapter introduction, you must add one.

Think of this as guiding a new reader through your work.

6. Figure and Table Captions

Every figure, plot, and table must include:

- A clear, descriptive caption
- An explanation of what the reader is viewing
- Why the figure or table matters

Use proper Quarto chunk options (e.g., `fig-cap`) to ensure captions render correctly.

There should be no unlabeled figures in your book.

7. Final Reflection Chapter

Your final chapter must be a reflective essay (minimum 400 words) addressing:

- What you learned this semester
- What challenged you the most
- What you are most proud of

- How your skills and confidence have changed
- How reproducibility has influenced your thinking

This chapter should be written in thoughtful, professional prose.

Book Quality Standards

Your book should:

- Render cleanly from beginning to end
- Have consistent formatting across chapters
- Include clear headings and subheadings
- Avoid redundant code or messy output
- Suppress unnecessary warnings/messages
- Demonstrate attention to detail

This is not a collection of assignments.

It is a cohesive analytical portfolio.

Publishing Requirement

You must publish your completed Quarto book.

Your submission must include:

- A link to your published book
- Your complete Quarto book project folder

Your published book should be accessible and shareable.

Grading (100 Points Total)

- Structure & Organization (20 pts)
Proper chapter order, functioning book structure
 - Reproducibility (20 pts)
Clean rendering, working code, no errors
 - Presentation & Formatting (20 pts)
Consistent style, captions, polished layout
 - Chapter Introductions (10 pts)
Clear context provided for each chapter
 - Final Reflection (15 pts)
Depth, thoughtfulness, clarity
 - Professionalism (15 pts)
Overall quality, cohesion, publication readiness
-

Reproducibility Practice

Your book must:

- Render entirely from source files
- Load all required packages explicitly
- Avoid hard-coded outputs
- Avoid manual edits to results
- Use consistent chunk labeling
- Be reproducible on another machine

The goal is that anyone could download your project and recreate your entire book.

Takeaway

By completing this assignment, you will leave the course with:

- A complete analytical book
- A published portfolio artifact
- Demonstrated mastery of reproducible workflows
- Authorship credit
- A powerful representation of your growth

This is the culmination of your semester's work.

Make it something you are proud to share.

References

References

- Martinez, Christian. 2026. *nycOpenData: Convenient Access to NYC Open Data API Endpoints*. <https://martinezc1.github.io/nycOpenData/>.
- Spinu, Vitalie, Garrett Golemund, and Hadley Wickham. 2024. *Lubridate: Make Dealing with Dates a Little Easier*. <https://lubridate.tidyverse.org>.
- Wickham, Hadley. 2023. *Tidyverse: Easily Install and Load the Tidyverse*. <https://tidyverse.tidyverse.org>.